

Open Data Fusion for Approximate True Cost Accounting using Knowledge Graphs

Abstract

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1. Introduction

Neglecting costs which affect third parties in economic transactions leads to incorrect price equilibria. This not only reduces market efficiency but incentivizes unsustainable practices, by effectively subsidizing behavior that has hard-to-assess consequences.

To address this, one would have to consider (and assign costs to) all externalities relevant for the production of a good. Attempts at this difficult and time consuming practice are often called True Cost Accounting (TCA).

To illustrate what this looks like in practice, we may examine an existing report on the true price of coffee beans. For the specific brand of beans, [1] estimates the true cost at around 9.33€/ kg compared to the market price of 8€/ kg. The breakdown of the additional cost into categories can be seen in Figure 1. In Section 2 we revisit this example with our proposed methodology.

This work investigates whether it is possible to automate this process at scale by aggregating and processing public data to arrive at an approximation of the true cost. To this end we construct a temporal knowledge graph and categorize data sources into a predefined ontology. Each category is assigned a base unit and associated cost. Translating all metrics to costs yields a hidden cost per company, which, when put in relation to the revenue, allows for the very rough estimation of a true cost by product. We term this approach Approximate True Cost Accounting (ATCA) and provide a proof-of-concept implementation to facilitate further research and promote adoption of similar frameworks.

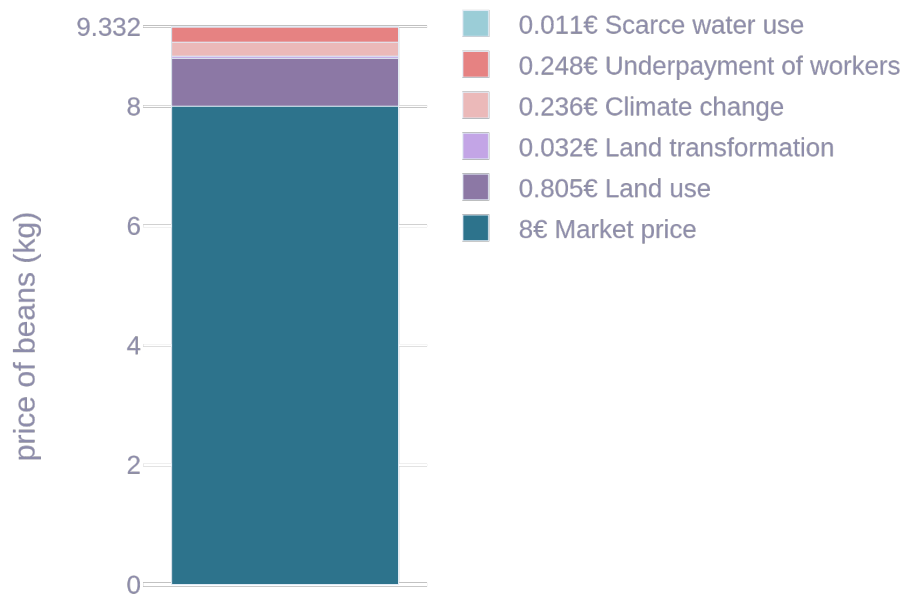


Figure 1: Example true cost calculation for coffee beans.
Data from [1].

1.1. Related Work

Several works exist, which explore the concept of pricing displaced harm with varying methodologies and target audiences.

The True Price Foundation [2] standardized and open-sourced a method for calculating a true price, which has been applied to different products such as [3], [4], [5], [1].

TEEBAgriFood [6] is a division of the The Economics of Ecosystems and Biodiversity (TEEB) initiative, which itself is part of the United Nations Environment Programme (UNEP). It currently spans 14 countries and prices natural factors that are usually ignored in an attempt to influence policy making.

The TCA Accelerator [7] is a network that advocates for the use of TCA together with the Global Alliance for the Future of Food [8], focusing on interdisciplinary knowledge sharing.

These and similar initiatives tackle the core problem of pricing externalities, but to the best of our knowledge, no prior work addresses the problem of automating this process end-to-end from existing data. In addition, they disproportionately focus on agriculture, while our approach works for products regardless of industry.

2. Methodology

Figure 1 presented the results of a proper true cost report by [2]. Imagine how we might approach this same product without the ability to inspect the production chain or ask the company for additional information. Instead we rely on the incomplete data we have. For a given year we may know the revenue of *Bocca Coffee* to be \$6.9 million. For illustrative reasons, imagine we also know the emissions to be 100 tons of CO_2 equivalent, alongside a water consumption of $6000m^3$ and electricity consumption $100Mwh$. We may then consult our table of per unit costs¹ to find the priced externalities from greenhouse gas emissions² at $312\text{€} / \text{ton}$ and water at $1.62\text{€} / m^3$. For electricity we use the estimate by [9] at $70\text{€} / Mwh$. The result of aggregating these externalities and breaking them down proportionally to a kilogram of beans can be seen in Figure 2.

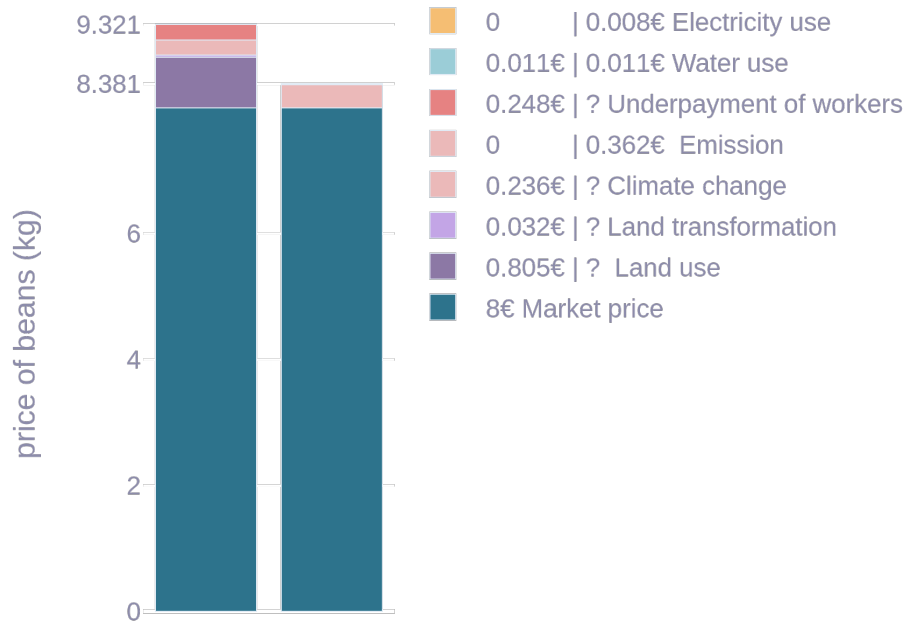


Figure 2: Example ATCA result for Bocca Coffee Beans (left) compared to real data (right).

The rest of this chapter explains the infrastructure needed to perform this approximation at scale. At the core of this work lies the KG, which the platform surrounding it attempts to make easily more accessible. This section is therefore divided into graph construction and graph utilization.

¹For example obtained using the methodology of [2]

²sometimes abbreviated Social Cost of Carbon (SCC)

2.1. Graph Construction

The data ingestion phase is designed to be flexible and extensible to serve different use cases. Inclusion of sources can be toggled in the interface. With the currently available sources graph construction and indexing is a matter of minutes on consumer hardware. This is subject to change with the inclusion of more or bottlenecked sources, such as those only available via Application Programming Interface (API). After adding the schema (see Appendix A), each source is parsed into triples and stored using the graph database oxigraph [10].

The available sources are described in the following.

2.1.1. Source: Wikirate

Wikirate [11] is an open data platform, which crowdsources and hosts a variety of company data relating to Environmental, Social and Governance (ESG) issues. Each data point regardless of type (Company, Metric, Project, Tag, Image, ...) lives in a shared namespace as so called cards, which can be fetched from an API.

To include Wikirate in the graph, a subset³ of 1000 companies with most data points associated is selected, alongside the 1000 most bookmarked metrics. These are cached and transformed from tabular format into triples, conforming to the schema described in Appendix A.

2.1.2. Source: OpenFoodFacts & OpenProductsFacts

OpenFoodFacts [12] a crowdsourced database of food products which tracks information such as brand, ingredients, nutritional value and allergens. The project was started in 2012 and expanded into several domains since then. Most notably, in 2018 OpenProductsFacts [13] was launched, which generalizes the venture to products, which do not fall under the more specialized categories such as food or beauty products. At the time of writing [12] includes around 4.4 million food articles, while [13] includes around 40.000 products. Both are exported once and formatted to triples. For prospective deployment, the respective APIs would have to be integrated to keep the data up to date.

2.1.3. Harmonization

With a diverse set of potential sources and in-source inconsistencies in data representation comes the burden of harmonizing the data, such that parts which reference the same conceptual entity are not disconnected.

Deciding on common vocabulary at data ingestion and enforcing the harmonized form throughout the graph lifetime ensures a baseline of interoperability.

Alongside cleanup of minor data quality issues, the most impactful harmonization steps in the pipeline are company name unification and metric categorization.

To avoid ambiguity we introduce IDs for each Company. To obtain a map from all possible representations of company name to its ID, we draw inspiration from the Natural Language Processing (NLP) technique of stemming to reduce a name down to a base form. Using regular expressions we strip away organizational indicators such as “inc.” or “limited” and non-ASCII characters. We apply this to the name of the company and any known aliases and store the reduced forms in a lookup table mapping to the created ID. Any collisions are recorded and either decided manually or discarded for the colliding aliases. Access to the lookup table is then provided by checking for the equally sanitized form of the query term. Conveniently the keys of the table can be used to generate company name completions in the user interface.

³The subset selection is done for scoping reasons in the proof-of-concept implementation. [11] has a long tail of largely isolated and therefore less interesting data points.

Metrics are also assigned an ID, however this is not enough to disambiguate. Metrics such as “Emission Scope 1” and “Scope 1 Emissions” clearly measure the same concept, but may have been collected by different parties or using different methodology. It is therefore necessary to find a mapping between arbitrary metrics with name, value and unit to a known category as part of an ontology. The proof-of-concept implementation uses a curated keyword matching approach for this. At the time of writing this only assigns a category to around 30% of metrics. There is a ceiling to the number of meaningful categories, as not every metric can be translated directly into a monetary value, however it is reasonable to believe that there is much room for improvement before this ceiling is reached. One potential improvement would be querying an Large Language Model (LLM) with the metric title and the task of assigning a category from the ontology. This approach is not error proof either, but with descriptive category names it is reasonable to believe it would perform and especially generalize better than a curated list of keywords.

Having described the data ingestion and processing pipeline we conclude this section with a hint for how to utilize existing resources.

Resource Description Framework (RDF) supports federated queries through the SERVICE keyword. This can be used to combine results from external endpoints, so long as there exists overlap between the graphs. For example, since companies can be identified by their OpenCorporates ID which is often present in the companies Wikidata representation. Therefore information from Wikidata can directly be integrated at query time, without modifications to the graph.

2.2. Graph Utilization

This chapter describes the way the **KG** is made accessible to a user. Besides the included user-interface, a qlever endpoint is exposed which provides syntax highlighting, completion and execution analysis via the qlever-ui. Lastly, to accommodate users unfamiliar with SPARQL Protocol and RDF Query Language (SPARQL), querying in natural language is supported through LLM-based translation, directly from the command-line interface.

2.2.1. UI

The user interface provides the core functionality for interacting with the **KG**. It first presents the user with the option to build the graph from a selection of sources, alongside a dropdown preview of relevant parts of the schema to help with orientation. Figure 3 shows this page.

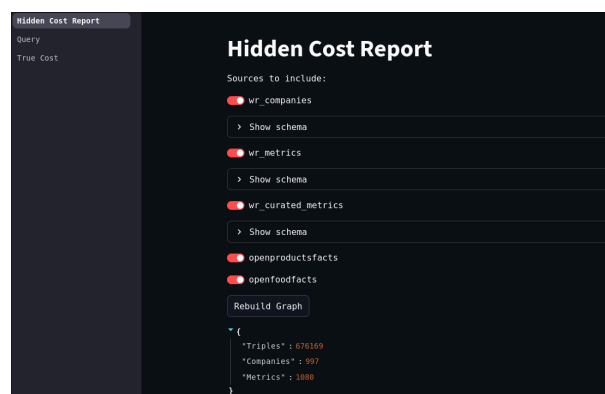


Figure 3: Front page of the UI.

The next tab, seen in Figure 4 allows for direct query access of the graph. It includes a search bar for quick retrieval of Company IDs and a text field for query design. Once submitted, the query is evaluated and the result is shown as a table and can additionally be downloaded in csv format. Notably, the access is read-only, exclusively allowing SELECT statements to prevent corruption of the **KG** once fully constructed and indexed.

Upon first open, the query tab shows an example query, selecting all metrics for the selected company. This demonstrates the schema applied in **SPARQL**, provides a convenient namespace and demonstrates that graph construction was successful.

company selection

apple

company id: C1578

query

```
PREFIX hcr: <http://hiddenreport.org/schema>
SELECT ?company ?metric ?value ?year WHERE {
  <id>= hcr:Name ?company .
  <id>= hcr:hasMetric ?obs .
  ?obs hcr:Value ?value .
  ?obs hcr:Year ?year .
  ?obs hcr:MetricID ?metricID .
  ?metricID hcr:MetricTitle ?metric .
}
```

submit

company	metric	value
0 Apple Inc.	MSA statement signed	Signed - no
1 Apple Inc.	Female Worker Fatalities (G4-L46-a)	Unknown
2 Apple Inc.	Meets Minimum MSA Requirements	Yes
3 Apple Inc.	Indirect greenhouse gas (GHG) emissions (Scope 2), GRI 305-2 (formerly G4-EN16-a)	378790
4 Apple Inc.	MSA statement signed	Signed by D
5 Apple Inc.	Comprehensive Income Net of Tax	369670000
6 Apple Inc.	CS: I.S.B. Personal Data Protection Policy	No
7 Apple Inc.	Hazardous Air Pollutant (HAP) emissions, GRI 305-7-a (formerly G4-EN21-a)	Unknown
8 Apple Inc.	Annual Revenue	215635000
9 Apple Inc.	Meets Minimum UK MSA Requirements	No

Figure 4: Query page of the UI with an example query.

Lastly, the UI includes a tab for computing the true cost of a product given a company and year. Once company and year are selected, the user is shown a breakdown of the metrics that were assigned a category with monetary equivalent. They are filterable by supercategory and aggregated to arrive at a total hidden cost estimate for the company in the given year.

Since the OpenProductsFacts dataset does not yet include a satisfactory level of coverage, the user is instead asked to provide just the price of a product of interest, which acts as a placeholder until OpenProductsFacts is well populated and integrated. Given this, the revenue is retrieved and the hidden cost per dollar of revenue is computed to obtain the proportional true cost of the product of interest.

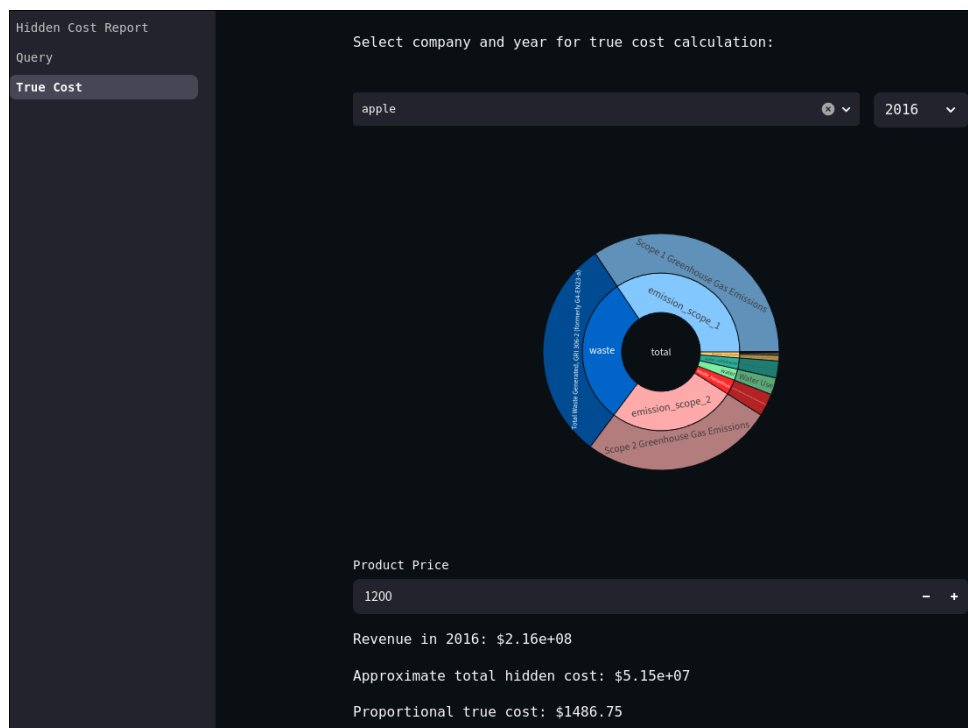


Figure 5: True cost page of the UI with an example selection.

For more general queries to the knowledge base we refer to the Qlever UI, which is explained in the following.

2.2.2. Qlever

Qlever [14] is a query engine for **RDF** using the **SPARQL** language. It provides a user interface that has many quality of life features, such as autocompletion, syntax highlighting, detailed error messages and query performance profiling to help with debugging. The Command Line Interface (CLI) provides a command to index the graph with Qlever and start up its server and ui, which can then be opened and used in a browser. For users who prefer to forego manual query writing entirely, we also offer a natural language to **SPARQL** interface:

2.2.3. Query Translation

Translation of natural language is accomplished by zero-shot prompting an **LLM** with a rough description of the schema, along with instructions to avoid empirically observed mistakes. The full preprompt can be found in Appendix B. The model tasked with this translation is a degree of freedom, during development the model “deepseek-r1-distill-llama-70b” hosted by the KISSKI initiative was used, as it delivered decent reasoning capability while being cheap at inference, though a more capable model may be selected for better results. Due to the number of potential queries and nondeterministic responses, quantitative assessment of said results proved difficult. We instead offer qualitative remarks:

For simple retrieval the approach is mostly reliable, but it begins to struggle with nested queries or complex dependencies. For example for the question “find companies that increased their emissions without loss of revenue in 2016” in one instance produced the **SPARQL** query:

```
SELECT ?company ?emissionValue2016 ?revenueValue2016
WHERE {
    ?company hcr:hasMetric ?emission2016 ;
             hcr:hasMetric ?revenue2016 ;
             hcr:hasMetric ?emission2015 ;
             hcr:hasMetric ?revenue2015 .

    ?emission2016 hcr:MetricID ?emissionMetricID ;
                 hcr:Year 2016 ;
                 hcr:Value ?emissionValue2016 .

    ?revenue2016 hcr:MetricID ?revenueMetricID ;
                 hcr:Year 2016 ;
                 hcr:Value ?revenueValue2016 .

    ?emission2015 hcr:MetricID ?emissionMetricID ;
                 hcr:Year 2015 ;
                 hcr:Value ?emissionValue2015 .

    ?revenue2015 hcr:MetricID ?revenueMetricID ;
                 hcr:Year 2015 ;
                 hcr:Value ?revenueValue2015 .

    ?emissionMetricID hcr:MetricCategory "emission" ;
                     hcr:MetricID ?emissionMetricID .

    ?revenueMetricID hcr:MetricCategory "revenue" ;
                     hcr:MetricID ?revenueMetricID .

    FILTER (?emissionValue2016 > ?emissionValue2015 && ?revenueValue2016 >= ?
```



```
revenueValue2015)
}
```

This is logically and syntactically correct, but since it contains self references of the metric IDs, which the **SPARQL** engine interprets as the condition that a triple `metricID hasMetric metricID` exists, it returns no results. One way to address these trivial errors would be an agentic setting, where the model can continually query the **KG** until the query has results or it has been verified that the required information is not present in the graph. We leave this as future work.

3. Evaluation

Evaluation of the system can be approached from several directions. One can measure the initial target of approximating true cost, by comparing output to existing true cost report. Secondly one can examine the graph on its own and apply traditional **KG** quality measures. Lastly one can qualitative judge the performance of the **LLM** query translation component, as already explored in the previous chapter.

3.1. True Cost Accuracy

Many existing **TCA** reports ([4], [6], [15], [3], [5], [1]) focus on a single product. Unfortunately the lack of data granularity makes this difficult to reproduce. As a rough approximation we compute the hidden costs per dollar of revenue for a company and multiply this with the price of a product to get the hidden cost. This approach has several assumptions which do not hold in practice:

- All products contribute to all metrics in proportion to their price.
- The entire supply chain is operated by the same company⁴
- The costs associated with metrics are location independent and constant.

Figure 2 demonstrates that an approximation using our methodology may at least be on the same order of magnitude as a diligent true cost report, yielding a true cost gap of 0.36€ (4.76%) compared to the 1.32€ (16.5%) found by [1]. while highlighting some of the shortcomings, such as the fact that the approximation is more of a lower-bounding. The scarce data landscape, both in the realm of true cost reports as well available supply chain data makes a rigorous assessment of our systems accuracy difficult.

3.2. KG Assessment

KG quality has several dimensions, the exact definitions and distinctions being subject of dispute in the literature [16]. The importance of each dimension, depends on the usecase. We follow the methodology of [16] and assess the **KG** along the axes of accuracy, completeness, consistency, timeliness, trustworthiness and availability.

3.2.1. Accuracy

Accuracy describes the degree of factuality as well degree of conformity to the specification in our case, **RDF** (syntactic validity). [16] Quantifying the accuracy requires ground truth data, which is unavailable by design. If it were available for parts of the graph, it would simply be incorporated and serve at most as a lower bound to accuracy. Syntactic validity on the other hand is guaranteed by the current set of data ingestion methods, as incorrectly formatted data is discarded to preserve validity. This is subject to change with integration of less well formatted data sources.

3.2.2. Completeness

Completeness describes how much of the data required for a particular task is present in the graph. [16] The application at hand theoretically requires complete knowledge of the universe and one

⁴There are inconsistencies between metrics, scope 2 and 3 emissions for example, account for external factors, while water usage only accounts for in house consumption.

could state that technically the completeness of the KG is exactly 0. However for practical purpose we limit ourselves to obtainable data and instead model companies and metrics as a bipartite graph. Connecting each company to the metrics we have about it and computing the edge density gives us a measure that can meaningfully substitute completeness. At the time of writing, this density lies at 12.32% across the time of recording. When broken down to yearly level the density drops to 1.02%. A density this low, even after filtering for the most documented companies on wikirate, is alarming and severely limits the amount of companies that meaningful true price approximations can be made for.

To get a better picture, Figure 6 shows the head of the distribution of around 250 companies, while Figure 7 shows the opposite perspective: the number of companies that report a value for each metric.

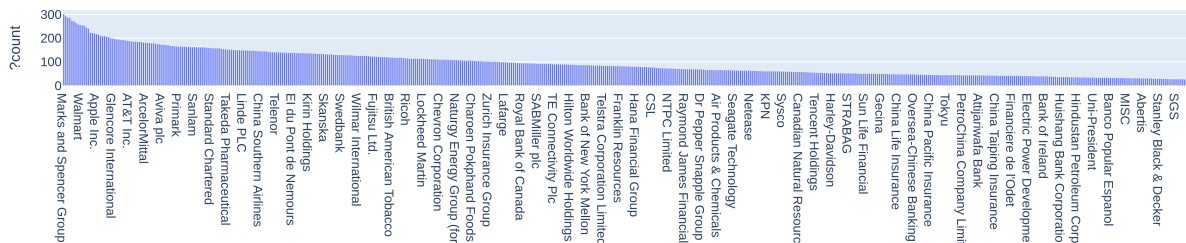


Figure 6: Number of Metrics per Company.

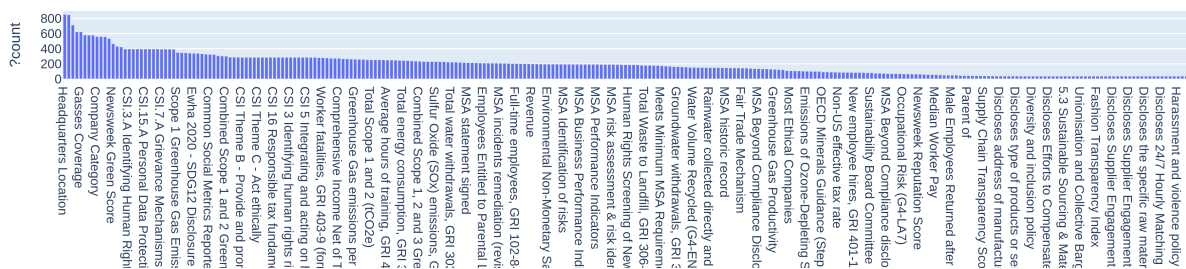


Figure 7: Number of Companies per Metric.

3.2.3. Consistency

Consistency is a measure of in-graph contradictions. [16] While some contradictions are to be expected due to the crowd sourced nature of the data landscape, a certain degree of consistency can be enforced through the ontology. For example, since the relationship between the metrics “emissions scope 1”, “emissions scope 2” and “emissions scope 1 and 2 combined” is obvious and (once encoded in the ontology) a violation of this relationship could trigger self correction or alert to the need of intervention. Since this behavior would depend on a complete and robust ontology we leave it as future work.

3.2.4. Timeliness

Timeliness describes whether the data in the graph is up-to-date with respect to the task. [16] For the core task of ATCA the bulk of the data, that is metrics and their values, fortunately does not grow stale, as it is strictly scoped to the year it was recorded for. It is rare that metric values change long after collection. Company structure and product lineup do evolve. however the core mutates slowly enough, that outdated data is not of great concern. The limitation instead lies in data collection upstream and with how recently it was fetched from there. High timeliness is not of critical importance to the task of ATCA, nonetheless it is advised to limit attempts to a span of at least several years and at most several decades back in time to strike a balance between data abundance and staleness.

3.2.5. Trustworthiness

Trustworthiness describes how much the data can be trusted as objective and verifiable. [16] It strongly correlated with, but distinct from accuracy in the sense that it has a subjective component in judgement of authority and credibility for data sources. The KG constructed in this work has partial trustworthiness, since it includes crowdsourced and self reported data. While often hard to validate, the data is unlikely to be completely false, as companies are likely to face consequences when consistently misreporting.

4. Discussion

This work attempted to explore whether it is possible to automatically calculate a holistic true price of any product across all industries just from publicly available metrics about the company that produces it. At the time of writing it unfortunately has to be concluded, that this is the not the case. Even for true price lower bounding and when tolerating the clearly violated assumptions the resulting values are simply too unreliable (often with volatility across orders of magnitude year over year) to be regarded as little more than noise.

Nonetheless there is hope for this type of approach in the future. With more careful scoping, diligent reporting on the side of companies and inclusion of curated data it might well be possible to create a tool to inform decision-making both along a supply chain and for individual consumers.

Acknowledgements

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5. Appendix

A Graph Schema

```
@prefix : <http://hiddencostreport.org/schema#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix dc: <http://purl.org/dc/elements/1.1/> .

@prefix wr: <https://wikirate.org> .
```

```
:Company a rdfs:Class .
:Product a rdfs:Class .
:Article a rdfs:Class .
:Metric a rdfs:Class .
:Observation a rdfs:Class .
:Dataset a rdfs:Class .
:Datapoint a rdfs:Class .
:Product a rdfs:Class .

:Name a rdf:property ;
    rdfs:range rdfs:literal .

:OpenCorporatesID a rdf:property ;
    rdfs:domain :Company ;
    rdfs:range rdfs:literal .
```

```

:MetricTitle a rdf:property ;
  rdfs:domain :Metric ;
  rdfs:range rdfs:literal .

:MetricDesigner a rdf:property ;
  rdfs:domain :Metric ;
  rdfs:range rdfs:literal .

:Questions a rdf:property ;
  rdfs:domain :Metric ;
  rdfs:range rdfs:literal .

:ValueType a rdf:property ;
  rdfs:domain :Metric ;
  rdfs:range rdfs:literal .

:Unit a rdf:property ;
  rdfs:domain :Metric ;
  rdfs:range rdfs:literal .

:hasMetric a rdf:property;
  rdfs:domain :Company ;
  rdfs:range :Observation .

:Brand a rdf:property ;
  rdfs:domain :Product .

:Ingredient a rdf:property ;
  rdfs:domain :Product .

:ManufacturedIn a rdf:property ;
  rdfs:domain :Product .

```

B Query Translation Prompt

Your task is to translate natural language questions into SPARQL queries.

The knowledge base is comprised of several sources, most prominently wikirate metrics.

The sources are not all harmonized, avoid specific vocabulary (tons might be called tonnes elsewhere).

The schema of the knowledge graph is as follows:

PREFIX hcr: <http://hiddencostreport.org/schema#>

```

hcr:company_id hcr:Name companyName ;
  hcr:OpenCorporatesID openCorporatesID ;
  hcr:hasMetric _:metricObservation .
_:metricObservation hcr:MetricID metricID ;
  hcr:Value observationValue ;
  hcr:Year observationYear ;
hcr:MetricID hcr:MetricTitle metricTitle ;
  hcr:MetricDesigner MetricDesigner ;
  hcr:Unit unit ;
  hcr:ValueType valuetype ;
  hcr:MetricType metricType ;
  hcr:MetricCategory MetricCategory ;

```

```
hcr:Questions questions ;
```

Refer to metrics by their category, the categories are:

derived

disclosure_rate

disclosure_single

electricity_consumption

emission

emission_scope_1

emission_scope_12

emission_scope_123

emission_scope_13

emission_scope_2

emission_scope_23

emission_scope_3

revenue

unmapped

waste

waste_hazardous

waste_hazardous_recycled

waste_nonhazardous

waste_nonhazardous_recycled

waste_recycled

water

water_recycled

water_withdrawal

Return a single code block denoted with ```.

Only use the provided predicates.

Include PREFIX in your query.

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